

Be My Anesthesiologist:

A Virtual Anesthesia Volunteer Network for Remote Care

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I. BACKGROUND

Rural hospitals and military medical units frequently face a critical shortage of anesthesiologists. In the United States alone, projections indicate a deficit of over 7,000 anesthesiologists by 2033, with shortages particularly severe in rural areas [1]. Many rural hospitals rely entirely on Certified Registered Nurse Anesthetists (CRNAs) without any on-site anesthesiologists. In some states, CRNAs are the sole anesthesia providers in nearly all rural hospitals [1]. Similarly, frontline military field hospitals or combat surgical teams may lack a dedicated anesthesiologist, placing anesthesia care in the hands of CRNAs or general physicians working under extreme conditions [2]. These gaps lead to delays in urgent surgeries, limited pain management options, and elevated stress for on-site providers. Although telemedicine has begun to address some of these challenges through institutional tele-anesthesia programs [2], existing models are often rigid and unsuitable for smaller hospitals or ad-hoc emergencies at unconventional hours. There is a pressing need for a more agile, human-centered solution that can bring expert guidance to the bedside when it is needed most.

IV. CONCLUSION

This project aims to empower local clinicians, reduce unnecessary patient transfers, and ultimately save lives by extending anesthesiology expertise to places that need it most. The system allows anesthesiologists to participate on their terms while recognizing their contribution through salary bonus, CME credits, public acknowledgments, or charitable donations. Most importantly, it offers a way to create a lasting impact from anywhere in the world. This innovation has the potential to support rural health workers, military medics, and humanitarian responders who often work in isolation. By blending human generosity with emerging technologies, it creates a model that could be adapted for other specialties in the future.

III. RESULTS

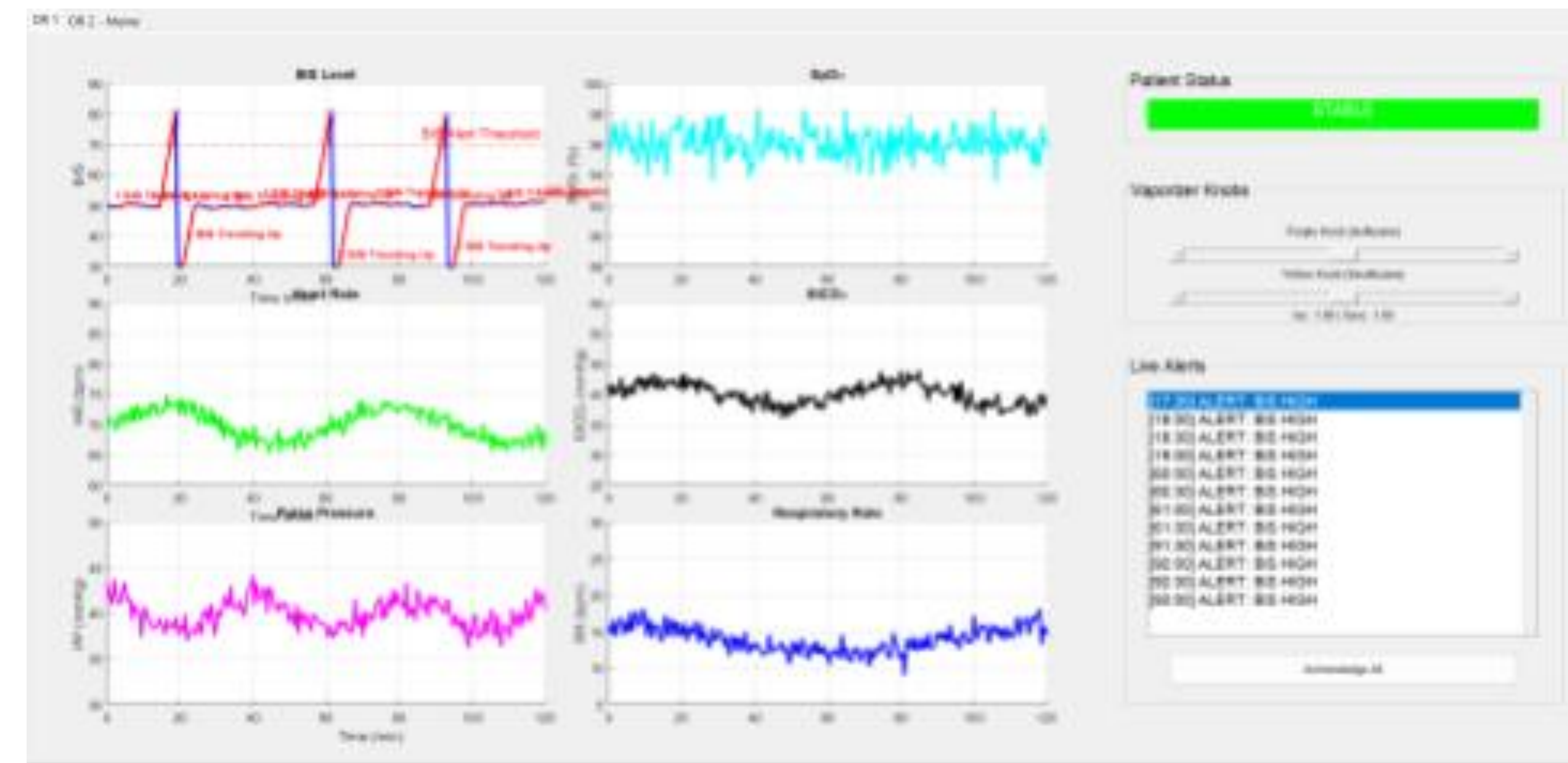


Fig. 1: Prototype AI Anesthesia Dashboard showing two hours of patient's vitals

Be My Anesthesiologist's AI dashboard provides a dynamic, real-time visualization of patient vitals in a simulated operating room environment. As shown in Figure 1, the interface continuously plots six key parameters: BIS level, heart rate, pulse pressure, SpO_2 , end-tidal CO_2 , and respiratory rate on separate axes for intuitive trend assessment. Abnormal BIS values trigger automated alerts that populate the "Live Alerts" panel. A patient status indicator ("STABLE" or "ALERT") provides a quick-glance summary, while interactive vaporizer sliders allow adjustment of isoflurane and sevoflurane concentrations. The dashboard features a tabbed interface for monitoring multiple operating rooms simultaneously; tab labels (e.g., "OR 2 – Maine") reflecting the geographic distribution of surgical sites, demonstrating how a single platform could enable anesthesiologists to supervise rural hospitals across the country. This responsive UI mirrors the complexity of real clinical environments and illustrates how an AI-enhanced monitoring and alerting system can support safe care delivery, especially when paired with remote anesthesiologist oversight.

Currently, vaporizer adjustments are achieved through hardware integration: the software communicates with a microcontroller (e.g., Arduino) that controls two servo motors. These motors are equipped with custom 3D-printed claws designed to grip and turn the physical vaporizer knobs on standard anesthesia machines. Future iterations will aim to eliminate the need for hardware by integrating directly with anesthesia systems through a proprietary digital interface, pending patent development.

II. METHODS / SYSTEM DESIGN

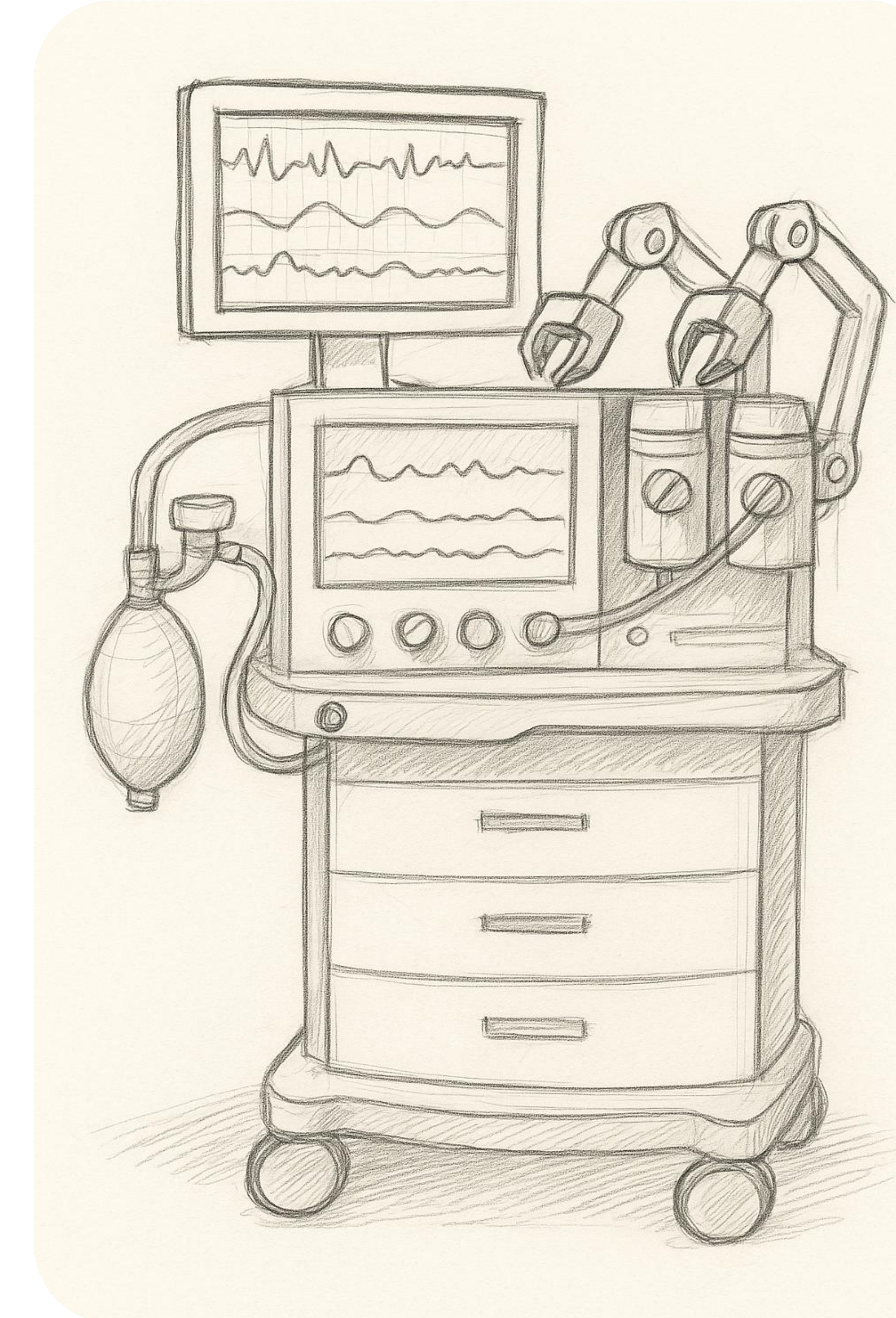


Fig. 2: Early-stage prototype sketch illustrating AI-assisted integration with an anesthesia workstation. The concept features servo-driven arms for automated vaporizer adjustments, a simplified vital monitoring interface, and a modular setup reflecting the system's research-stage development.

Concept Overview

Be My Anesthesiologist is modeled after *Be My Eyes*, which connects blind users with sighted volunteers through live video [3]. This system builds a verified, volunteer network of credentialed anesthesiologists who provide real-time guidance to clinicians in rural or military settings. Each help request is broadcast to multiple volunteers, allowing the first available anesthesiologist to respond through secure video or audio. Participation requires formal credentialing, ensuring only certified professionals provide support. Through live connections, anesthesiologists can view monitors, assess patient status, and advise on airway management, drug administration, or crisis response, creating an agile, human-centered tele-anesthesia model that respects provider schedules and urgency.

AI Integration and Decision Support

Artificial intelligence functions as a triage and monitoring assistant, inspired by *Be My Eyes'* collaboration with OpenAI's GPT-5. The AI continuously evaluates patient data, detects urgent trends, and prioritizes cases for human volunteers. It can:

- Analyze live patient vitals (BIS level, heart rate, pulse pressure, SpO_2 , end-tidal CO_2 , and respiratory rate).
- Detect anomalies and trigger "ALERT" status.
- Recommend anesthetic adjustments based on patient characteristics [4].
- Auto-document key interventions to reduce clinician workload [5].

If no anesthesiologist is immediately available, the AI offers protocol-based stabilization guidance until human support joins. This hybrid system blends AI's analytical power with human expertise to enhance safety and situational awareness in high-stress environments.

V. REFERENCES & ACKNOWLEDGES

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